



College of Engineering, Construction and Living Sciences  
Bachelor of Information Technology  
ID721001: Mobile Application Development  
Level 7, Credits 15  
**Project**

## **Assessment Overview**

In this **individual** assessment, you will develop two mobile applications using **React Native** and **Expo**.

## **Learning Outcomes**

At the successful completion of this course, learners will be able to:

1. Implement and publish complete, non-trivial, industry-standard mobile applications following sound architectural and code-quality standards.
2. Identify relevant use cases for a mobile computing scenario and incorporate them into an effective user experience design.
3. Follow industry standard software engineering practice in the design of mobile applications.

## Assessments

| Assessment | Weighting | Due Date                         | Learning Outcome |
|------------|-----------|----------------------------------|------------------|
| Practical  | 20%       | Sunday 7th September at 11.59 PM | 1, 2             |
| Project    | 80%       | Sunday 9th November at 11.59 PM) | 1, 2, 3          |

## Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday 9th November at 11.59 PM**.

## Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** over all assessments in **ID721001: Mobile Application Development**.

## Submission

You **must** submit all application files via **GitHub Classroom**. Here is the URL to the repository you will use for your submission – [https://classroom.github.com/a/Mq0MeQ\\_x](https://classroom.github.com/a/Mq0MeQ_x). If you do not have not one, create a **.gitignore** and add the ignored files in this resource - <https://raw.githubusercontent.com/github/gitignore/main/Node.gitignore>.

Create a branch called **project**. The latest application files in the **project** branch will be used to mark against the marking rubric. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **12.00 AM**.

## Authenticity

All parts of your submitted assessment **must** be completely your work. The use of **AI tools** is not permitted in this assessment. You need to demonstrate to the course lecturer that you can meet the learning outcomes for this assessment. You **must** complete the <https://moodle.op.ac.nz/mod/page/view.php?id=1178172> form and put this in the root of your repository. Failure to do this may result in a mark of **zero** for this assessment.

## Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic** policies. Learners can view policies on the **Otago Polytechnic** website located at <https://www.op.ac.nz/about-us/governance-and-management/policies>.

## Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

## Resits

Resits and reassessments are not applicable in **ID721001: Mobile Application Development**.

## Instructions

### Functionality - Learning Outcomes 1, 2 (20%)

- Application v1 with the following functionality:
  - Data is fetched from a local **JSON** file.
  - App icon and splash screen.
  - Styled using **Tailwind CSS**.
  - **Recipe Details Screen Features:**
    - \* Display the recipe's name, cuisine, image, ingredients, difficulty, prep time, cook time, servings, tags, nutrition, average rating and reviews.
    - \* Display a heart icon in the top-right corner. Tapping it adds or removes the recipe and saves the changes.
    - \* Display a plus icon next to the heart icon. Tapping it adds or removes the recipe's ingredients and saves the changes, ensuring no duplicate ingredients are stored.
  - **Daily Specials Screen Features:**
    - \* Display a list of five randomly selected recipes with difficulty level Easy or Medium.
    - \* Each item should show the recipe's name, cuisine, image and average rating.
    - \* Tapping an item should navigate to the **Recipe Details Screen**.
  - **All Recipes Screen Features:**
    - \* Display a list of all recipes.
    - \* Each item should show the recipe's name, image, difficulty level, total time and average rating.
    - \* Tapping an item should navigate to the **Recipe Details Screen**.
  - **Favourites Screen Features:**
    - \* Display a list of saved favourite recipes.
    - \* Each item should show the recipe's name, image and average rating.
    - \* Tapping an item should navigate to the **Recipe Details Screen**.
  - **Ingredients Screen Features:**
    - \* Display a list of saved ingredients grouped by category.
    - \* Each item should display the ingredient name, quantity and category.
    - \* Display a trash icon. Tapping it removes the ingredient and saves the changes.
- Application v2, which builds on v1 with the following functionality:
  - Select any four **Medium Features** from the list below:
    - \* Filter recipes by cuisine, tags, difficulty level and ingredient categories.
    - \* Search recipes by name, ingredient name or tag.
    - \* Sort recipes by prep time, cook time, total time, difficulty level or average rating.
    - \* Dark mode support with a theme toggle.
    - \* Delete all saved favourite recipes or ingredients.
    - \* Display recipes by times cooked or last cooked.
    - \* Display a list of the last five recipes viewed.
  - Select any four **Hard Features** from the list below:
    - \* Save and edit notes on recipes.
    - \* Add ratings and write reviews for recipes.
    - \* View combined ratings and user reviews.
    - \* Manage user profiles with preferences and dietary needs.
    - \* Add new recipes.

- \* Update and delete existing recipes.
- \* Meal planner to schedule recipes by day.
- Choose two features of your choice. These features should be different from the ones listed above.
- Create an **APK** for both v1 and v2.

### **Code Quality and Best Practices - Learning Outcomes 1, 3 (35%)**

- Write clean, readable, and maintainable code.
- Use modular, reusable components and avoid code duplication.
- Keep code simple and easy to understand.
- Ensure the app is responsive across different screen sizes and orientations.
- Optimise performance by managing resources efficiently.

### **Version Control - Learning Outcome 3 (5%)**

- Regular commits are made throughout development, demonstrating meaningful progress.
- Commit messages are clear and descriptive.
- The repository shows a clean and logical commit history that reflects the development process of the applications.

### **Testing - Learning Outcome 3 (10%)**

- Test the applications with three different users for each version (six users total), observing how they navigate and use features.
- Collect written feedback on usability issues, confusing UI and bugs.
- Write a brief report summarising the user testing, including:
  - Tester information (anonymised).
  - Tasks performed.
  - Positive feedback received.
  - Areas for improvement identified.
  - Bugs or errors found.
  - Suggestions on how feedback could be used to improve the applications (no requirement to implement).

### **Reflection - Learning Outcomes 1, 3 (5%)**

- Write a short reflection (300-500 words) on your development process and learning throughout the project, including:
  - Challenges you faced and how you overcame them.
  - What you would do differently in future projects.
  - New skills, tools or practices you learned during the project.

### **Presentation - Learning Outcome 2 (5%)**

- Record a 5-10 minute screen recording of your applications.
- Demonstrate the v1 and v2 features.
- Make sure the video is clear and easy to follow.
- Submit a link to the presentation in your repository's **README.md** file.